

introduction to

DATA DATA DATA
VISUALISATION

#1

↓
Exploratory

Declarative
↑

basics

Whats Is dataviz ?
Why do we visualize ?
Visual Functions

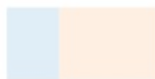
dataviz for data exploration

Search for
meaning using
visualisation

dataviz for communication

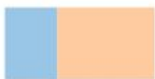
Show findings
using visualisation

Class & TP #1



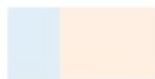
28 & 29 march

Class & TP #2



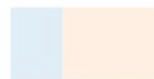
14 & 19 / 20 april

Class & TP #3



5 & 9 may

Class & TP #4



19 & 23 may

LAST CLASS & TP

- I. Whats is it ?
- II. Why do we visualize ?
- III. Visualisation Functions
- IV. Starting to explore with datavis

LAST DAY

- I. Whats is it ?
- II. Why do we visualize ?
- III. Visualisation Functions
- IV. Starting to explore with datavis

TODAY

- I. Quizz
- II. Chart of the day
- III. Try dashboards
- IV. About dashboards

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How many
variables can we
represent with this
chart type ?



How many
variables can we
represent with this
chart type ?



When do we use
those chart ?
To show what
function / relationship
in the data ?



Area



Stacked area

What difference
between those two
chart types ?

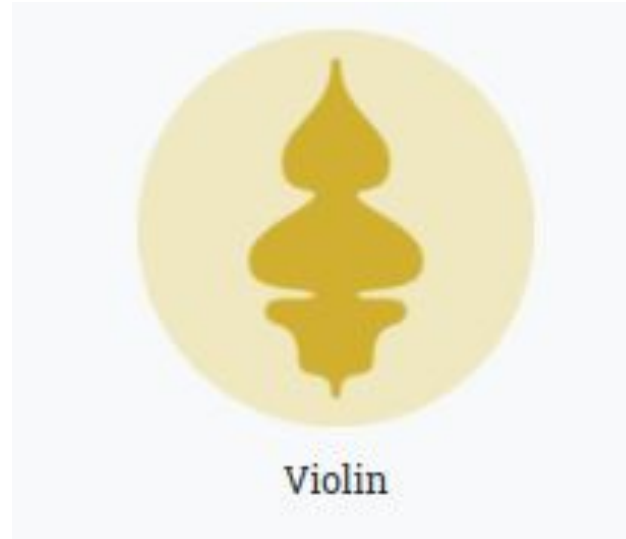


Stacked area



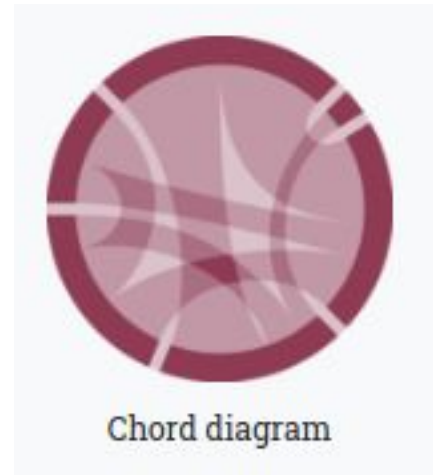
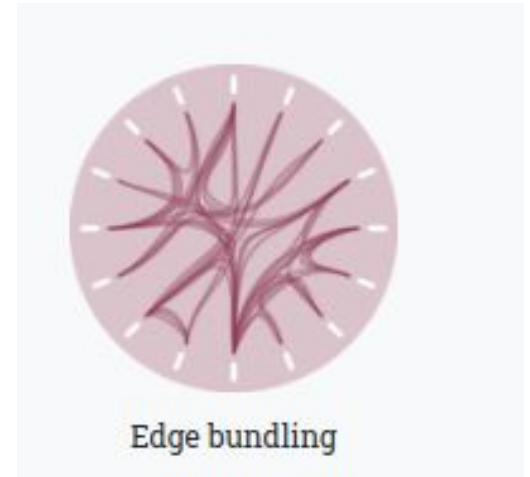
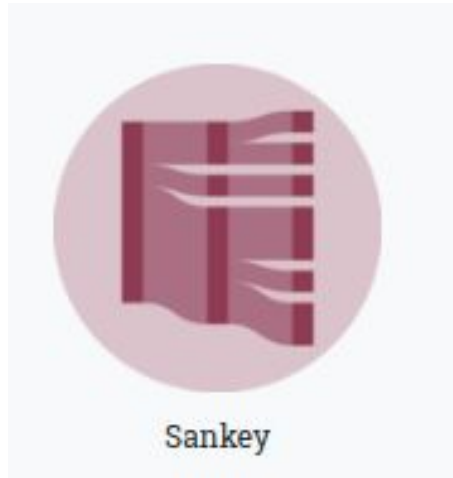
Streamchart

When do we use a violin plot ?



What are the differences between those chart types ?

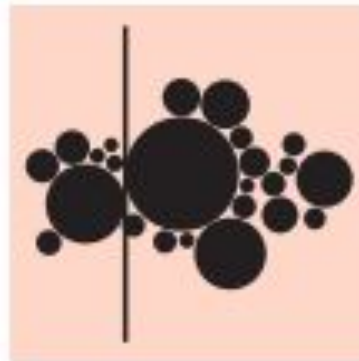
- What variables (number and type) ?
- What functions / relationship in the data ?



When do we use a beeswarm plot ?

- What variables (number and type) ?
- What functions / relationship in the data ?

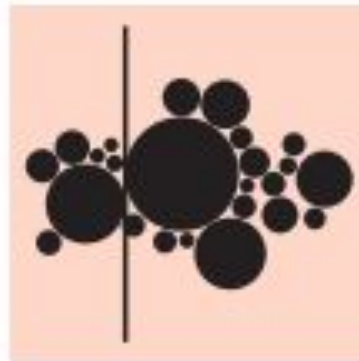
Beeswarm



When do we use a beeswarm plot ?

- What variables (number and type) ?
- What functions / relationship in the data ?

Beeswarm



Examples

<https://pudding.cool/projects/vocabulary/>

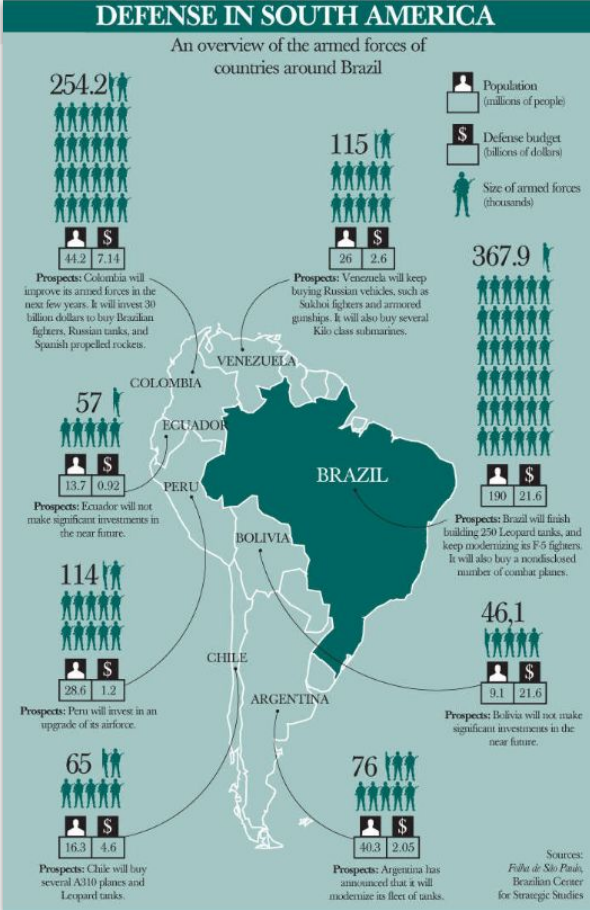
https://pdeuxa.github.io/TeamOc_HackViz21/

<https://www.nytimes.com/interactive/2020/04/02/us/coronavirus-social-distancing.html>

- I. Quizz
- II. Chart of the day
- III. Try dashboards
- IV. About dashboards

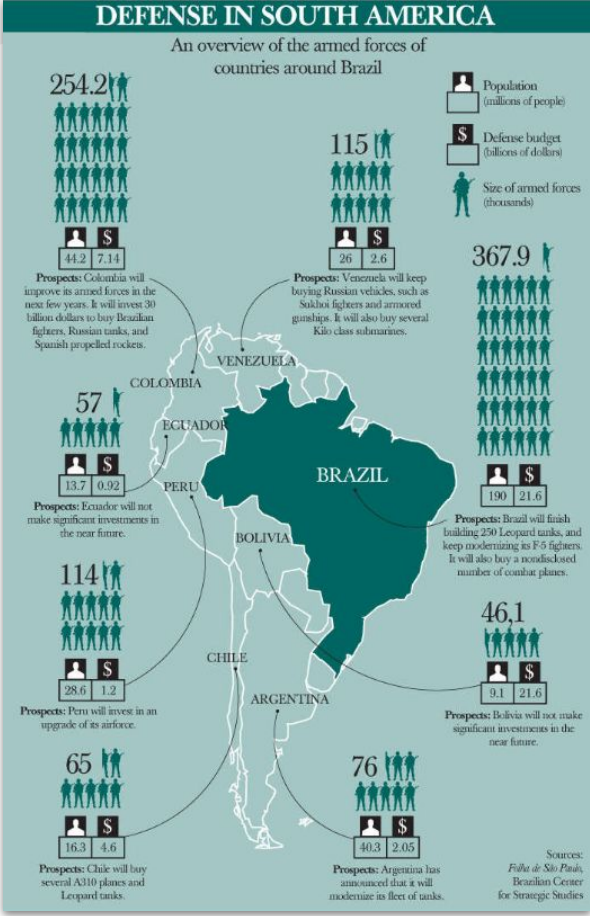
I. Chart of the day

#2



I. Chart of the day

#2



Deviation

Emphasise variations (+/-) from a fixed reference point. Typically the reference point is zero but it can also be a target or a long-term average. Can also be used to show sentiment (positive/neutral/negative).

Correlation

Show the relationship between two or more variables. Be mindful that, unless you tell them otherwise, many readers will assume the relationships you show them to be causal (i.e. one causes the other).

Change over Time

Give emphasis to changing trends. These can be short (intra-day) movements or extended series traversing decades or centuries. Choosing the correct time period is important to provide suitable context for the reader.

Part-to-whole

Show how a single entity can be broken down into its component elements. If the reader's interest is solely in the size of the components, consider a magnitude-type chart instead.

Ranking

Use where an item's position in an ordered list is more important than its absolute or relative value. Don't be afraid to highlight the points of interest.

Distribution

Show values in a dataset and how often they occur. The shape (or 'skew') of a distribution can be a memorable way of highlighting the lack of uniformity or equality in the data.

Magnitude

Show size comparisons. These can be relative (just being able to see larger/bigger) or absolute (need to see fine differences). Usually these show a 'counted' number (for example, barrels, dollars or people) rather than a calculated rate or per cent.

Spatial

Aside from locator maps only used when precise locations or geographical patterns in data are more important to the reader than anything else.

Flow

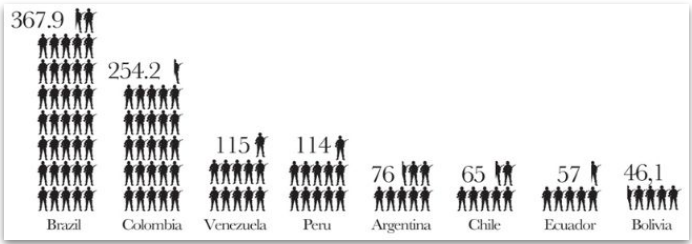
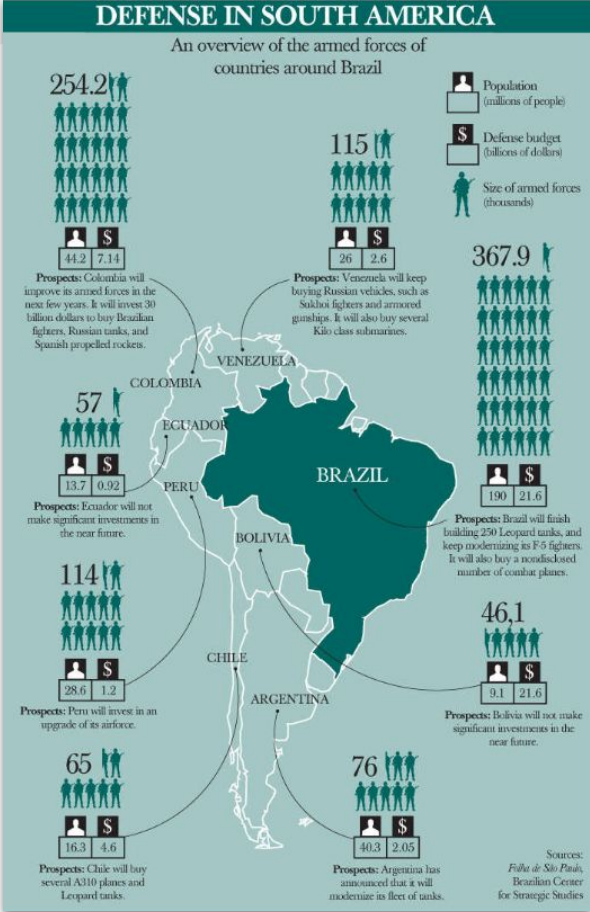
Show the reader volumes or intensity of movement between two or more states or conditions. These might be logical sequences or geographical locations.

Does this chart enable :

- to compare countries on variables ?
- to organize, to order countries ?
- to see links or correlation between variables ?

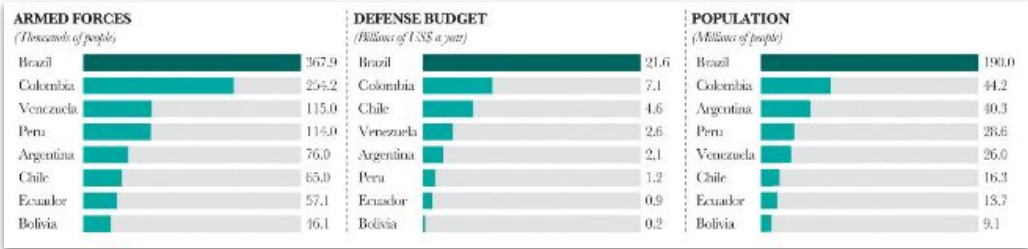
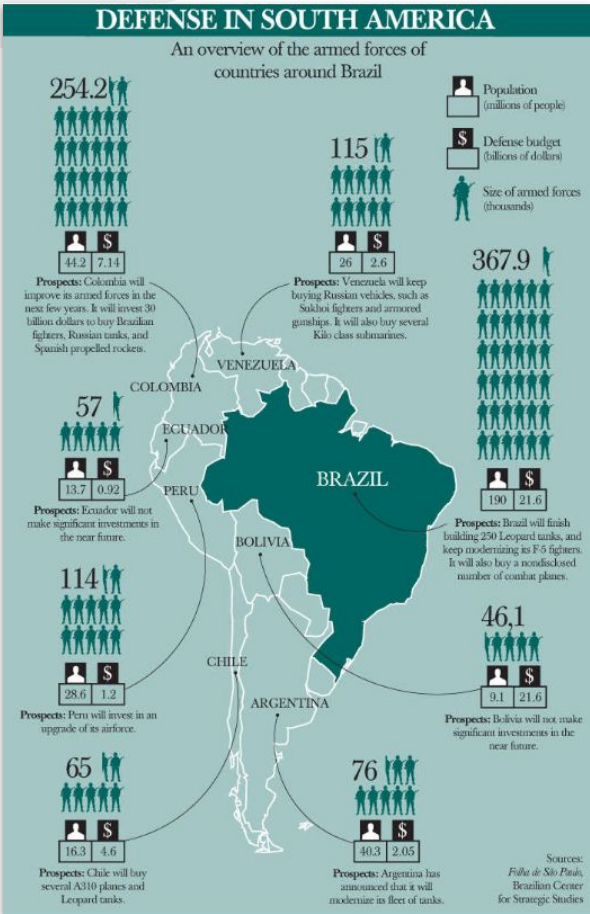
I. Chart of the day

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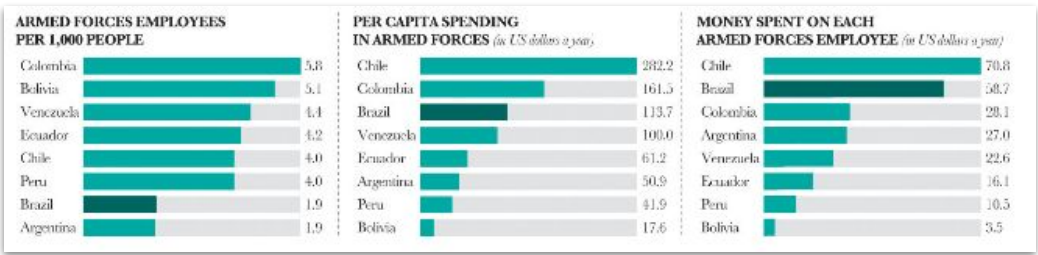
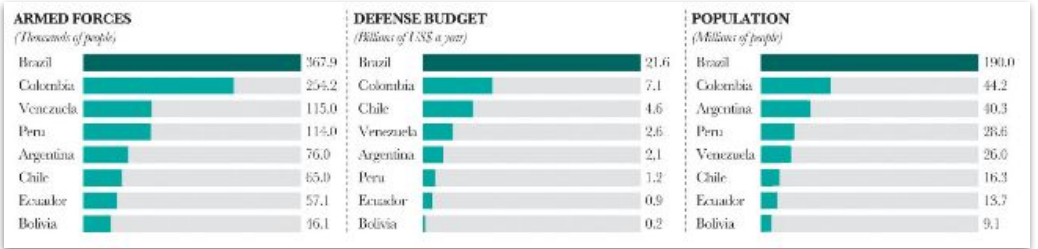
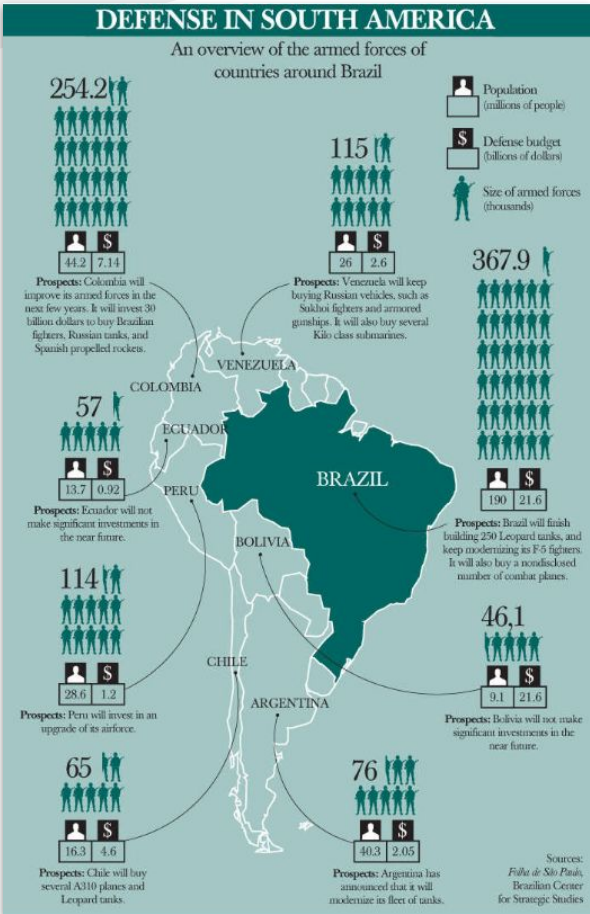
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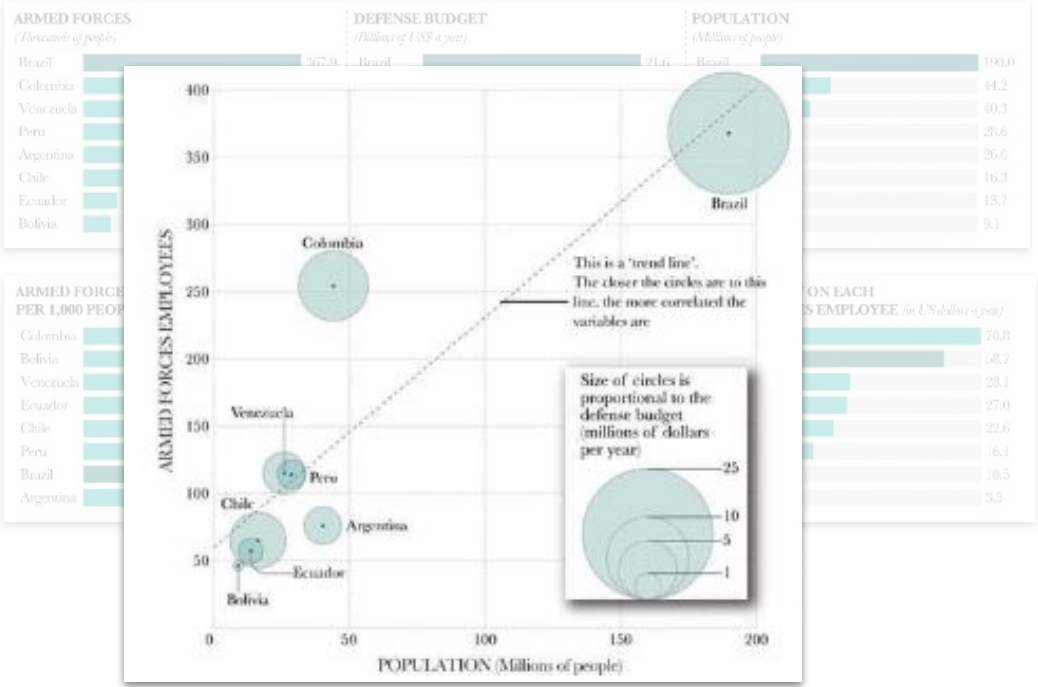
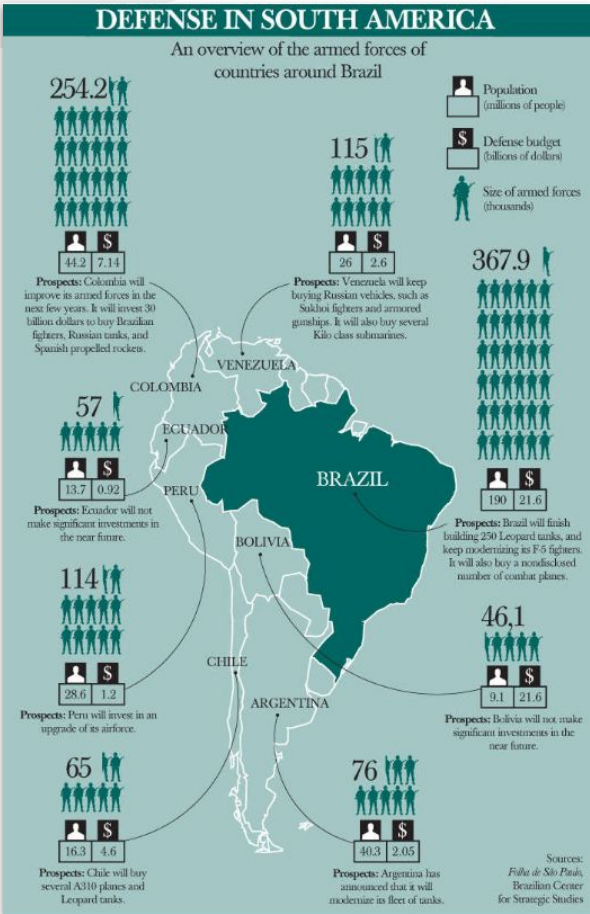
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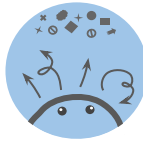


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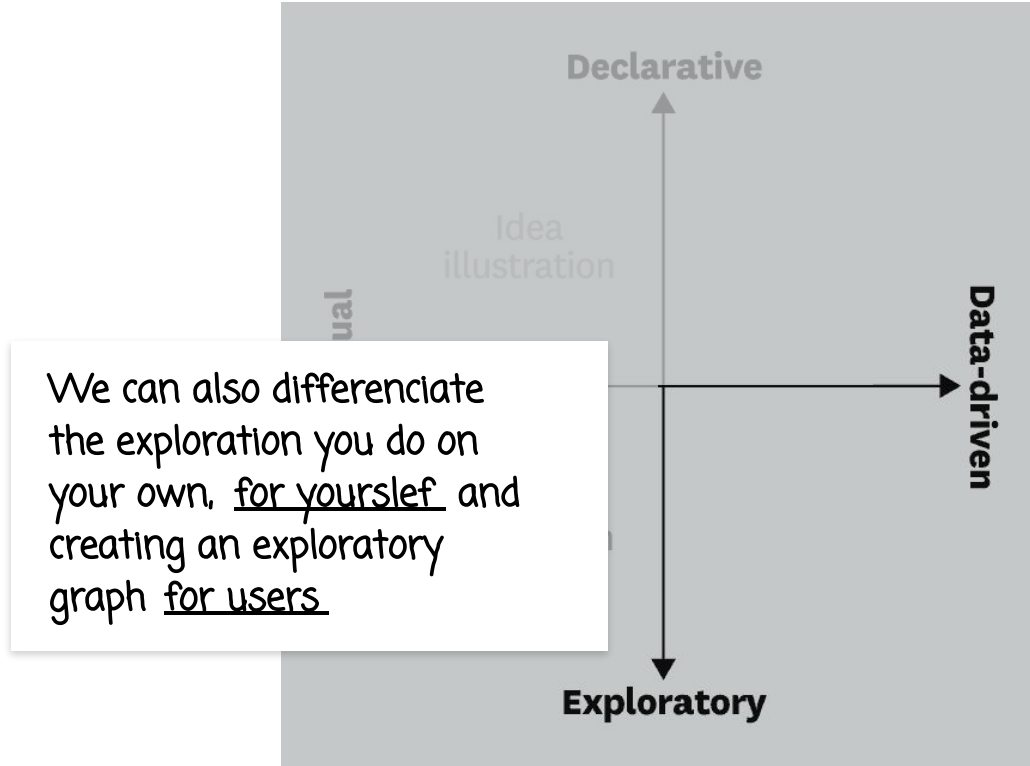


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#2

communicating
information



trying to figure something out

- confirmation
- discovery

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exploratory graph/tools for users

dashboard ?



Tableau Multichannel Attribution ...
windsor.ai



Hyper - Admin & Dashboard Template ...
themes.getbootstrap.com



Sales Performance Dashboard | Sales ...
boldbi.com



dashboard examples from real companies ...
geckboard.com



Sales Dashboards Template & KPI ...
clidata.com



Zoho Flow dashboard
zoho.com



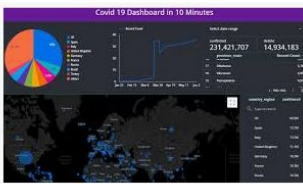
Dashboards
docs.datastax.co



Customer Service Insights dashboards ...
docs.microsoft.com



Dashboards For Project Manag...
clidata.com



How to Build a Stunning Interactive ...
towardsdatascience.com



Dashboard Templates For Admins ...
colorlib.com



data visualisation et storytelling ...
fanvoice.com



Pin on Data visualization



Financial Performance Dashboard ...



Investor Dashboard - Personal ...



Business Dashboards ...



SEO dashboard using Google Analytics ...



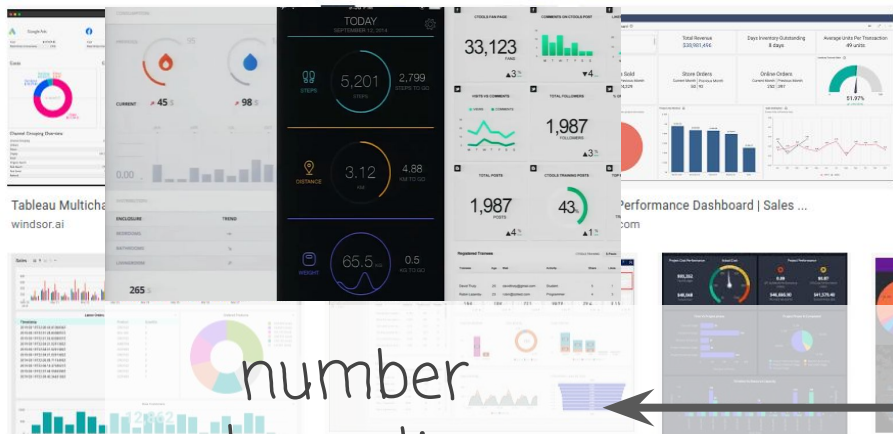
How to design a sleek dashboard UI | by ...

#2



exploratory graph/tools for users

dashboard ?



number decoration



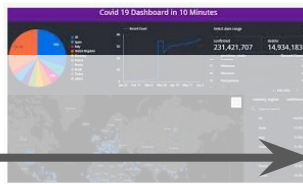
dashboard examples from real companies ...
geckoboard.com



Sales Dashboards Template & KPI ...
clidata.com



Zoho Flow dashboard
zoho.com



Dashboard templates for Admins
colorlib.com



data visualisation et storytelling ...
fanvoice.com



Pin on Data visualization



Financial Performance Dashboard ...



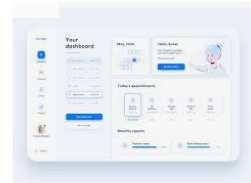
Investor Dashboard - Personal ...



Business Dashboards ...



SEO dashboard using Google Analytics ...



How to design a sleek dashboard UI | by ...

II. About dashboards

#2



Hi. My name is Moritz Stefaner.

I live and breathe data visualization —
as an independent designer and consultant.

I help organizations, researchers and businesses to
find truth and beauty in relevant and meaningful data.





#2

problematic

German train system is open, can but don't have to book

Goal : maximize use without having too full trains

Indirect ways to affect travelers #



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resources

Predictive models of passengers load but no tools for humans to work with that data

Many different tables in different places

Printing and drawing at peak time



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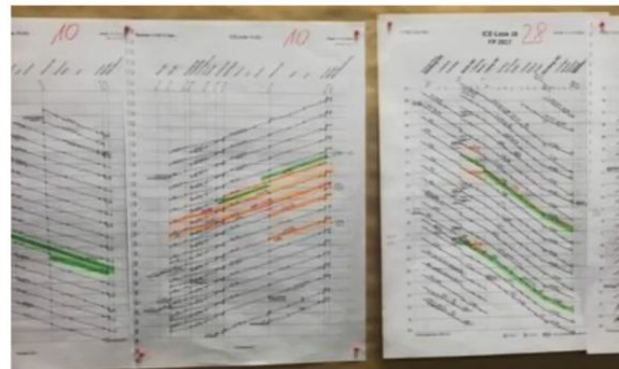
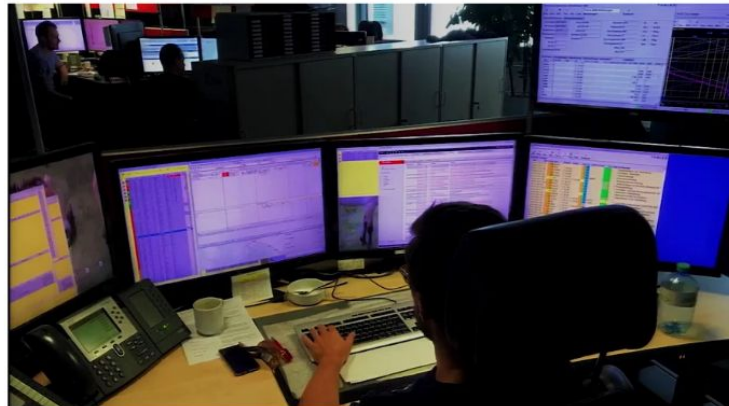
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process

Understand data and users





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Immersion (work with real data)

Prototyping with accessible tools (notebooks, tableau ...) for real use





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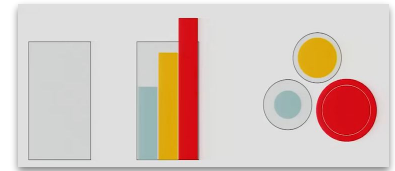
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Define a specific visual language to chose encodings





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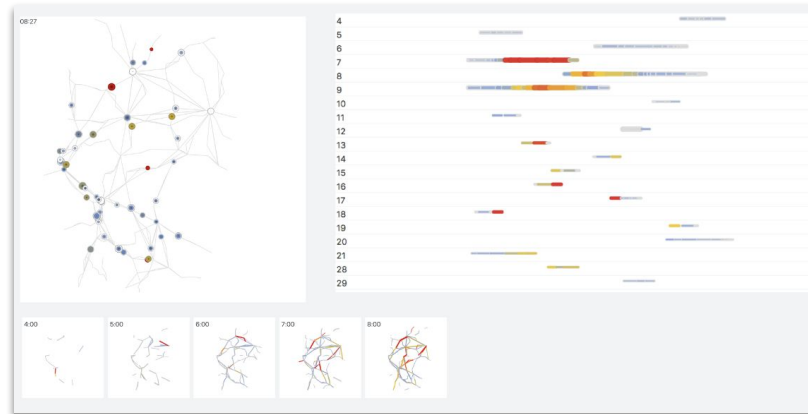
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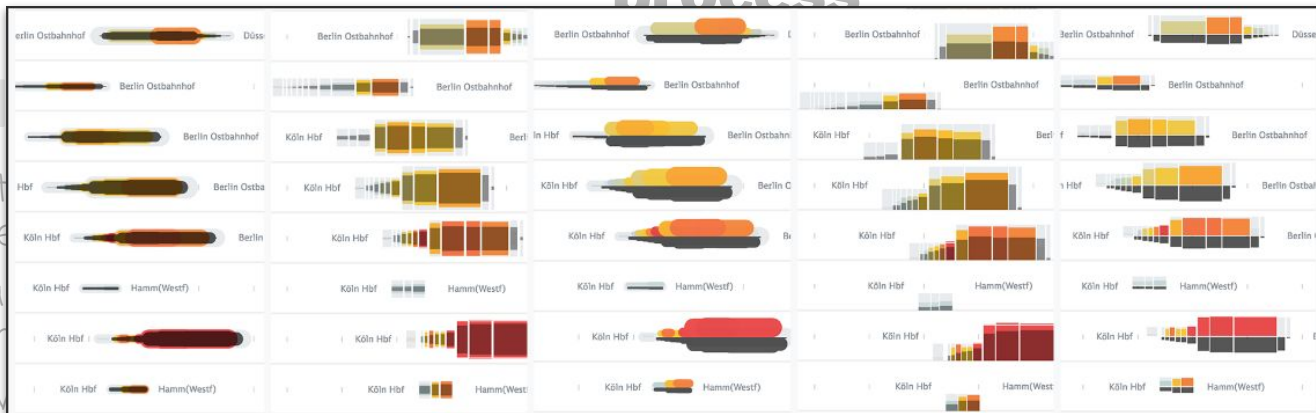
Favor quick explorations over perfectionism to find the right angle



II. About dashboards



process



problem

German train
don't have

Goal : make
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Indirect way

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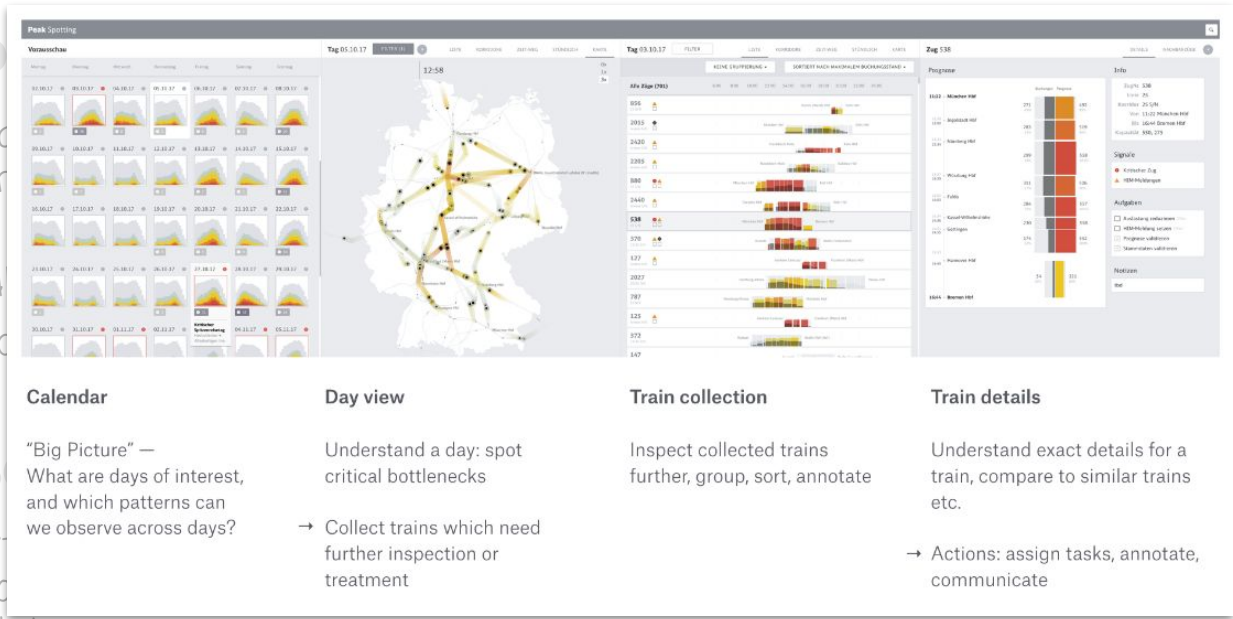
Favor quick explorations over
perfectionism to find the right angle

Doubt the data (missing ? do we
measure the right thing ?)

Test alternatives and iterate



process



Calendar

“Big Picture” —
What are days of interest,
and which patterns can
we observe across days?

Day view

Understand a day: spot
critical bottlenecks

→ Collect trains which need
further inspection or
treatment

Train collection

Inspect collected trains
further, group, sort, annotate

Train details

Understand exact details for a
train, compare to similar trains
etc.

→ Actions: assign tasks, annotate,
communicate

Refine and information architecture
(combining views) + UI

Visualize + analyse user behavior : what's
actual useful for users over time



#2

problematic

German train system is open
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Goal : maximize use without
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Indirect ways to affect travel

resources

Predictive models of passenger behavior
but no tools for humans to analyze
that data

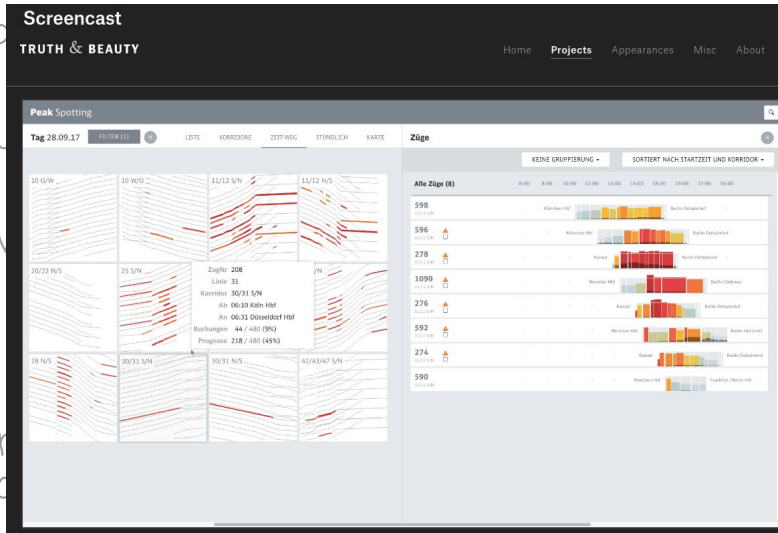
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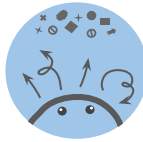
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**dataviz for data
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Search for
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**dataviz for
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Show findings
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II. About dashboards

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Declarative

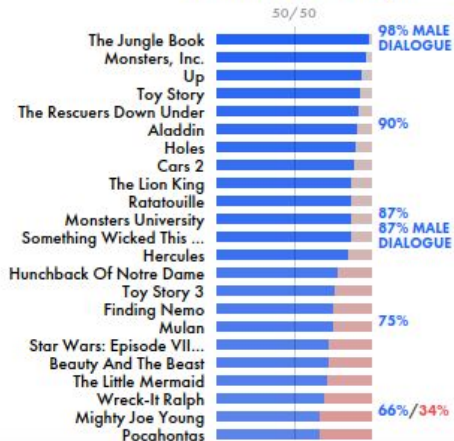


Screenplay Dialogue, Broken-down by Gender

2,000 Screenplays: Dialogue
Broken-down by Gender

Only High-Grossing Films: Ranked in
the Top 2,500 by US Box Office*

Men have 60%+ Dialogue



Gender Balance, +/- 10%



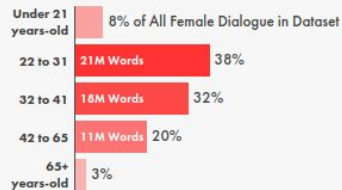
Women have 60%+ Dialogue



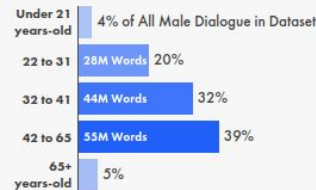
Percent of Dialogue by Actors' Age

AMONG 2,000 SCREENPLAYS, ALL GENRES, ALL YEARS

Actresses: Total Words, by Age

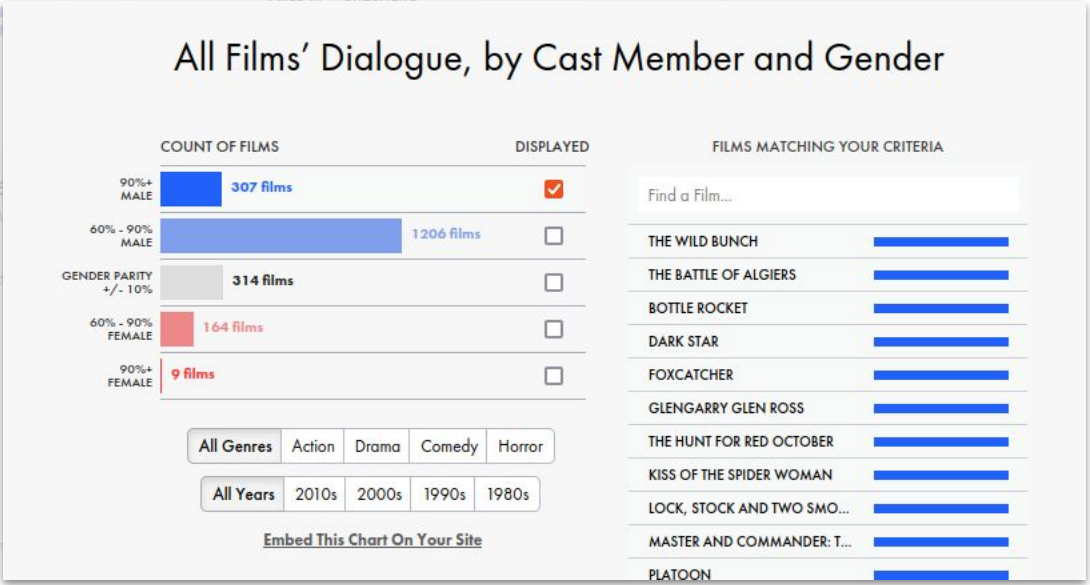
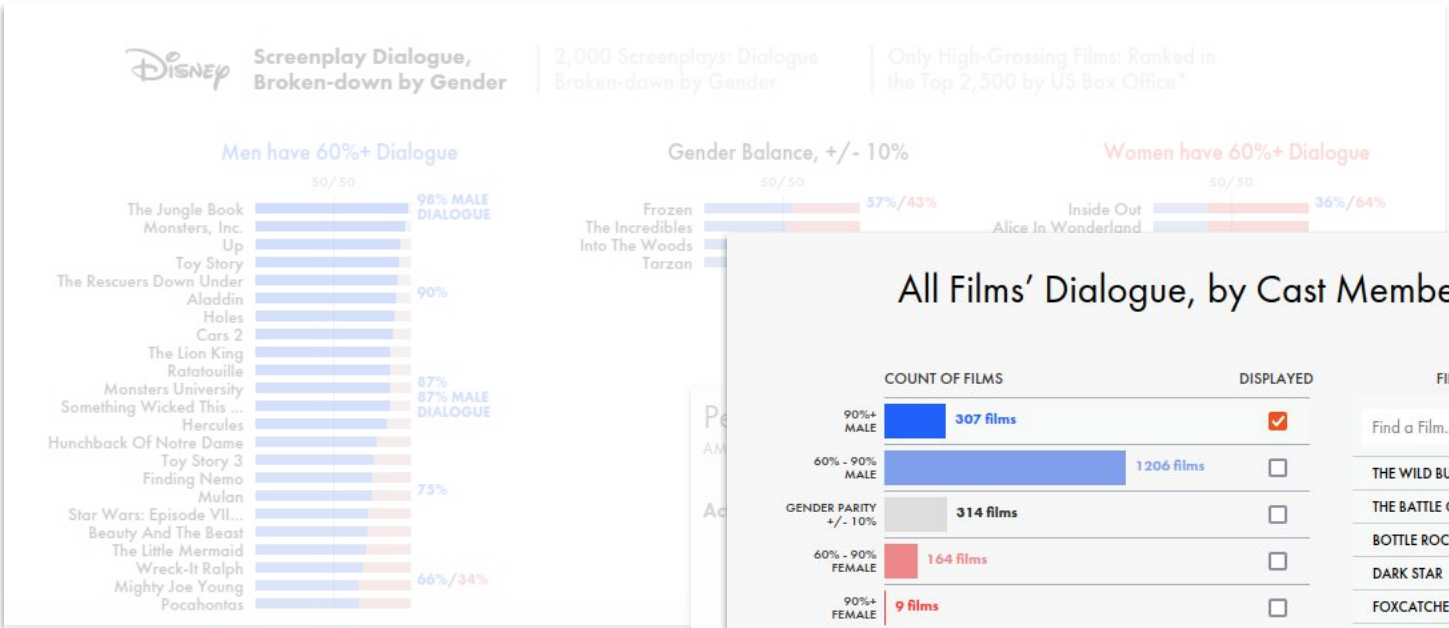


Actors: Total Words, by Age



Declarative

Exploratory





HOMEWORK

Overview

Skills :

Time : ~ 3 hours

Submission via Moodle

any kind of submission (text, drawing ..) and file (.doc, .pdf ...)

Practice

Preparing the final homework, a full visualisation project with exploration and presentation:

Step 1 - Choose your dataset

- [Kaggle datasets](#)
- [Data is plural](#)
- [Data.world](#)
- [Toulouse Dataviz Hackaviz datasets](#)

Step 2 - Context and data assessment

Step 3 - Explore the dataset